

Front Potted Interface Assembly



I know that Apehaenger has documented this very well in a kicad schematic that helped me out alot, so this has already been addressed before, but since I began digging into this before realizing others had solved it, I made some additional notes:

There is a potted front assembly that interfaces the mowers top cover STOP-button switches to the mainboard. This potted front assembly also interfaces the rain sensor for the mowers that have one (SA 650B ECO does not).

The potted front assembly has 3 connectors.

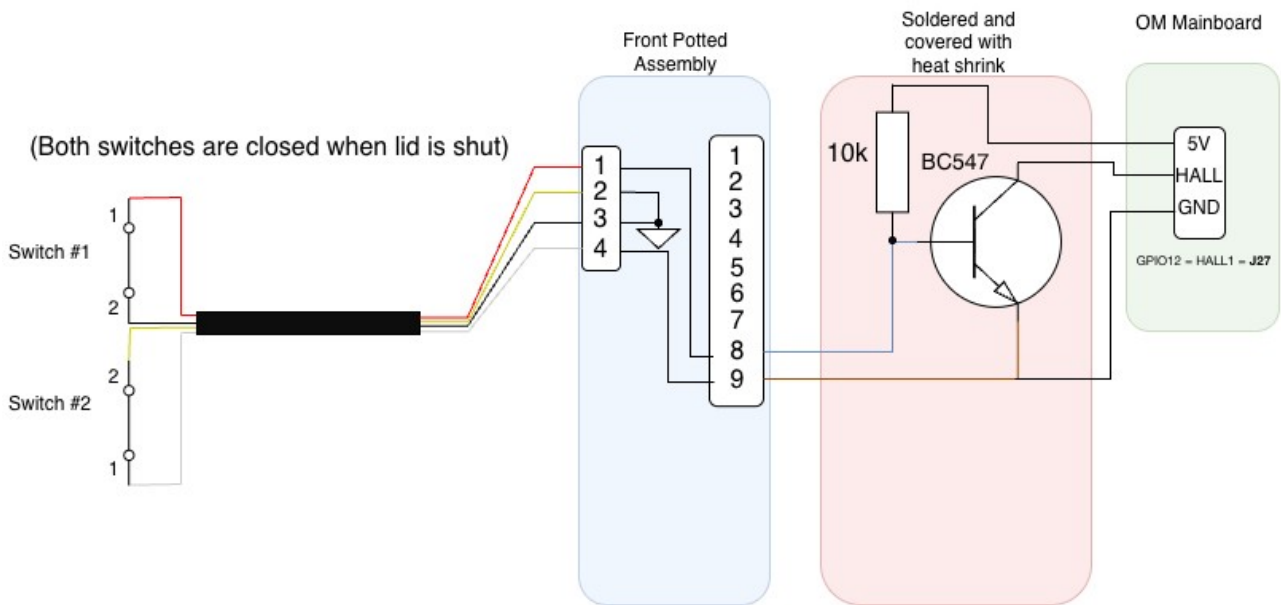
1x 9pin - Connected to the original mainboard

1x 4pin - Connected to top cover STOP-button switch

1x 2pin - Rain Sensor (NC for SA650, so I haven't drawn it in the block diagram / schematic)

A cable with 4 wires are routed from the mowers two top cover STOP-button switches. This cable goes to the front of the mower and is connected to the 4pin connector in the potted front assembly.

The colors in the overview indicate the colors where you'll have to do the reconnections. So the 9pin cable going from the front assembly and in towards the mainboard, the blue and brown cable on connector pin 8 and 9 go into the BJT base and emitter:

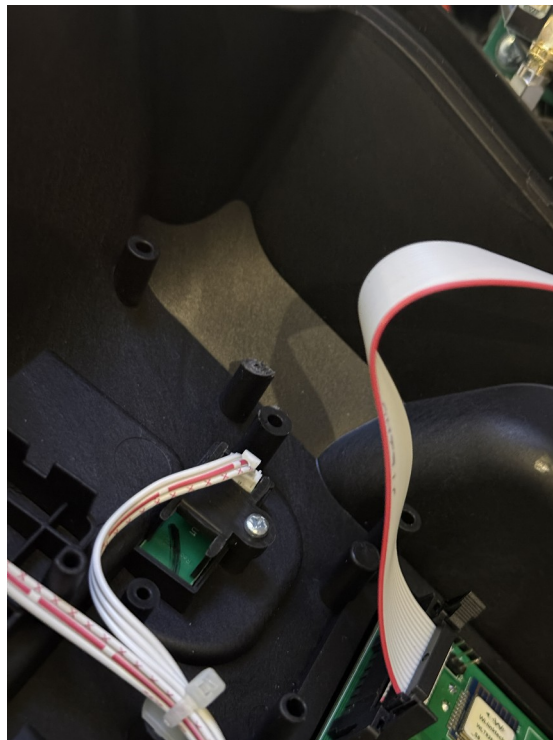


This allows us to use only one HALL input on the OM Mainboard for both of the STOP-button switches.

For the remaining 4x hall sensors on the mowers chassis (lift/tilt detection), I followed Apehaenger's schematic without making any notes myself. Schematic is attached to the discord message.

Modify chassis for better space

There is a plastic standoff on the inside of the top part of the mower housing that needs to be cut off. This standoff collides with the SMA connector on the GPS board that can cause it to fall off or even worse, break. By cutting this off, it seems to fit without any issues. Below is an image showing the standoff that should be snipped off.



Issues with CM4 flashing

I used a Raspberry Pi CM4 with eMMC and struggled a bit with getting it flashed. My issue was that usbboot was detecting the CM4 when powering on while holding the BOOT button, but it never showed up as a mass storage device on any PC i tried. I saw in the release history for USBBOOT that a release from 2021 added CM4 support, so I downloaded that and compiled it instead of the "latest release" I was using originally. This worked immediately, so download and compile this release if you have the same issues: <https://github.com/raspberrypi/usbboot/releases/tag/v2021.07.01>

Configuring xESC - WiFi vs Ethernet

When I was uploading the motor configurations to the drive/mower motors, I was connected to the OM mainboard via ethernet. This resulted in that no matter what xESC was exposed through the OM OS, only the rear left drive motor is found by VESC tool. Since I didn't use VESC tool to manually test-drive the motors while connected to them during this phase, i didn't realize until later that the specific motor configurations were not uploaded to the correct motors (mower motor config uploaded to drive motor). The symptom was that it could not drive in a straight line, drive motor was skipping alot, so luckily it was a very obvious system but took a while before I figured it out. So my tip to be sure is to configure over WiFi and manually drive the motors in VESC Tool to sanity check (Somewhat obvious warning but: **ensure to not do this safely wrt. the blades attached to the mower motor. Ideally do this without them installed**)